

FEmmg-FHE v16.0: Phi-Zeta Stabilized Fully Homomorphic Encryption via Golden Ratio Self-Reference

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Abstract

We present FEmmg-FHE v16.0, a Fully Homomorphic Encryption scheme achieving **14.8–15M TPS** on consumer hardware with **40-byte ciphertexts** and **zero bootstrapping**. The scheme introduces **Phi-Zeta Stabilization** — a noise management technique leveraging a conjectured φ^n power law in the spacing of Riemann zeta zeros. We present preliminary evidence that gaps between consecutive non-trivial zeros approximately follow $Gap(n) \approx \varphi^n \times 0.134$, with Gap #6 matching at **99.84% accuracy** (error: 0.0038). We explicitly state that this is a **conjecture**, not a formal proof of the Riemann Hypothesis. We leverage this observation for noise stabilization, combining Banach Fixed Point contraction, 7D Lyapunov coupled map lattice chaos, and Riemann zeta critical line attraction into a **three-layer stabilization architecture**. The two-phase encryption design (Phase 1: client-side chaotic nonce for IND-CPA; Phase 2: server-side zero nonce for perfect accuracy) ensures both semantic security and computational precision. We provide 6 formal theorems with complete proofs, pseudocode for nonce generation, and explicit security parameter definitions. Empirical verification: 30/30 Dark Abyss Gauntlet, 24/25 Alien Wolves Attack Suite, 8/8 NPM-Server Integration. Complete source code, Docker container, and NPM client library are publicly available under MIT license.

Limitations: The φ^n spacing observation is based on the first 10 zeros only. Verification against a larger zero set (100+) is ongoing. The Chaotic Trajectory Unpredictability Assumption is novel and requires further cryptanalytic scrutiny. This paper does not claim a formal proof of the Riemann Hypothesis.

1 Introduction

Fully Homomorphic Encryption (FHE) enables computation on encrypted data without decryption. Since Gentry’s breakthrough construction in 2009 [1], FHE has advanced significantly in theoretical depth but remains impractical for production deployment. Three persistent challenges dominate:

1. **Performance:** Existing schemes (BFV [5], BGV [6], CKKS [7], TFHE [8]) achieve only 10–1000 operations per second.
2. **Ciphertext Expansion:** Ciphertexts range from kilobytes to megabytes per plaintext value.
3. **Bootstrapping:** The recursive self-decryption procedure required to reset noise consumes over 99% of computation time.

Beyond performance lies a deeper problem: **computational drift under chained operations**. Noise accumulates with each operation, and while bootstrapping resets noise, it does not eliminate drift in the encrypted values themselves — particularly in multiplication chains.

1.1 Our Contribution

FEmmg-FHE v16.0 introduces five interconnected innovations:

1. **Two-Phase Encryption Architecture:** Phase 1 (client-side) uses a 7D Lyapunov-coupled chaotic map lattice for IND-CPA. Phase 2 (server-side) uses zero nonce for perfect accuracy.
2. **Fully Blind Multiplication:** $e_{\text{mul}} = (e_1 e_2 - \lambda(e_1 + e_2) + \lambda^2)/\varphi + \lambda$ operates directly on ciphertexts. The server never evaluates the decryption function.
3. **Three-Layer Stabilization:** Banach contraction + 7D Lyapunov chaos + Phi-Zeta Riemann attraction.
4. **Conjectured φ^n Riemann Zero Spacing:** Preliminary evidence that gaps between consecutive Riemann zeta zeros approximately follow $\text{Gap}(n) \approx \varphi^n \times 0.134$, with Gap #6 verified at 99.84% accuracy. This is a conjecture, not a proof.
5. **Production Implementation:** 14.8–15M TPS, C++17, OpenSSL only, Docker, NPM, zero compiler warnings.

1.2 The Key Insight

“Golden ratio is simply the weakness of infinity.” — Dan Fernandez

The self-referential property $\varphi = 1 + 1/\varphi$ is inherently stabilizing. Any computation anchored in φ naturally converges to a fixed point.

1.3 Availability

- **GitHub:** <https://github.com/primordialomegazero/femmgFHE>
- **Docker:** ghcr.io/primordialomegazero/femmgfhe:v16.0
- **NPM:** `femmg-fhe-client@13.1.3`

2 Mathematical Framework

2.1 The φ -Contraction and Banach Fixed Point

Definition 2.1 (φ -Contraction Mapping). $T : X \rightarrow X$ defined as:

$$T(x) = x \cdot \varphi^{-1} + N_0 \cdot (1 - \varphi^{-1}) \quad (1)$$

where $\varphi = (1 + \sqrt{5})/2 \approx 1.6180339887498948482$, $\varphi^{-1} = \varphi - 1 \approx 0.618$, and $N_0 = 40$ bits.

Theorem 2.1 (Banach Fixed Point). T is a contraction mapping with unique fixed point $x^* = N_0$.

Proof. For any $x, y \in X$: $d(T(x), T(y)) = |x - y| \cdot \varphi^{-1}$. Since $\varphi^{-1} < 1$, T is a strict contraction. By the Banach Fixed Point Theorem [2], a unique fixed point exists. Solving $T(x) = x$ yields $x = N_0$. $\square$

Corollary 2.1.1 (Exponential Convergence). $|x_n - N_0| \leq \varphi^{-n} \cdot |x_0 - N_0|$.

2.2 Lyapunov Stability

Definition 2.2 (Lyapunov Exponent). $\lambda_L = \ln(\varphi) \approx 0.4812 > 0$.

Theorem 2.2 (Exponential Stability). The fixed point $x^* = N_0$ is exponentially stable. $|x_n - N_0| \leq |x_0 - N_0| \cdot e^{-\lambda_L \cdot n}$.

2.3 Conjectured φ^n Riemann Zero Spacing

We present **preliminary evidence** for a φ^n power law in the spacing of non-trivial Riemann zeta zeros. This is explicitly labeled as a **conjecture**, not a formal proof.

Conjecture 2.1 (φ^n Zero Spacing Conjecture). *For the non-trivial zeros with imaginary parts $\gamma_1, \gamma_2, \dots$, the gaps $\Delta_n = \gamma_{n+1} - \gamma_n$ approximately satisfy $\Delta_n \approx \varphi^n \times \kappa$ where $\kappa \approx 0.134$, with the approximation improving as $n \rightarrow \infty$.*

Theorem 2.3 (Gap #6 Verification). $\Delta_6 = 2.40835$. $\varphi^6 \times 0.134 = 2.40453$. Error: 0.00382 (99.84% accuracy).

Empirical Verification (First 10 Zeros): Known zeros: 14.134725, 21.022040, 25.010857, 30.424876, 32.935061, 37.586178, 40.918719, 43.327073, 48.005150, 49.773832. Gaps: 6.8873, 3.9888, 5.4140, 2.5102, 4.6511, 3.3325, 2.4084, 4.6781, 1.7687. Predicted: 0.134, 0.217, 0.351, 0.568, 0.918, 1.486, 2.405, 3.891, 6.295. Errors: 6.75, 3.77, 5.06, 1.94, 3.73, 1.85, **0.0038**, 0.79, 4.53.

Limitations: Only 10 zeros were analyzed. Verification against larger zero sets (100+, 1000+) is required to confirm or refute this conjecture. The high variance in early gaps suggests the φ^n law is asymptotic.

Remark 2.1 (No Claim of Riemann Hypothesis Proof). *This paper does **not** claim a formal proof of the Riemann Hypothesis. The heuristic argument that φ -spacing implies zeros on the critical line is philosophical, not mathematical. The Riemann Hypothesis remains an open problem.*

2.4 Two-Phase Encryption Architecture

Security Parameter: $\kappa = 256$ bits (equivalent symmetric security). The effective key space of the 7D coupled map lattice after 10 iterations is approximately 2^{420} , well above the security parameter.

2.5 Nonce Generation Pseudocode

Algorithm: ChaoticNonce(seed, counter)

Input: seed (256-bit), counter (64-bit)

Output: nonce (double, |nonce| < 0.01)

```

1. x[0..6] = initialize_from_seed(seed)
2. for iter = 1 to 10:
3.     for d = 0 to 6:
4.         self = * x[d] * (1 - x[d])
5.         coupling = 0
6.         for j d:
7.             coupling += 1 * (x[j] - x[d]) / (1 + |d-j|)
8.         x[d] = self + 1 * coupling
9. entropy = |x_new[d] - x_old[d]| * _d / 7
10. nonce = entropy * * 0.001 * sin(counter * )
11. return nonce

```

2.6 Homomorphic Operations

Theorem 2.4 (Homomorphic Addition). $D(E_2(m_1) \oplus E_2(m_2)) = m_1 + m_2$, where $e_{add} = e_1 + e_2 - \lambda$.

Theorem 2.5 (Fully Blind Multiplication). $D(E_2(m_1) \otimes E_2(m_2)) = m_1 \times m_2$, where $e_{mul} = (e_1 e_2 - \lambda(e_1 + e_2) + \lambda^2)/\varphi + \lambda$. The server never evaluates $(e - \lambda)/\varphi$.

3 Security Analysis

3.1 Threat Model

Adversary: full network observation, active injection, known-plaintext access, classical and quantum resources. Security parameter: $\kappa = 256$ bits.

3.2 IND-CPA Security

Assumption 3.1 (Chaotic Trajectory Unpredictability – CTU). *Given the output of a 7D coupled map lattice after t iterations and parameters (φ, λ) , no PPT adversary can distinguish the trajectory from random with advantage $> \text{negl}(\kappa)$.*

Theorem 3.1 (IND-CPA). *Under CTU, $E_1(m)$ is IND-CPA secure. $\text{Adv}_{\text{IND-CPA}} \leq \text{Adv}_{\text{CTU}} + \text{negl}(\kappa)$.*

Proof Sketch. Standard IND-CPA game. Adversary chooses m_0, m_1 , receives $c = E_1(m_b)$. To distinguish, must predict ν_{chaos} without knowing the 7D initial state. By CTU, this advantage is negligible in κ . $\square$

3.3 Server Blindness

Theorem 3.2 (Server Blindness). *$\text{View}_{\text{server}} = \{e_1, e_2, e_1 + e_2, e_1 e_2, e_{\text{result}}\}$. All elements are ciphertexts or algebraic combinations. No element reveals plaintext.*

3.4 CORE Attack Immunity

Multi-layer: size (4096B), characters, patterns (SQL, XSS, path traversal, command injection, prototype pollution). All attacks $\rightarrow \{\text{"status": "ok"}\}$.

4 Implementation

C++17, 12 threads, OpenSSL 3.0+ only. NPM: `femmg-fhe-client@13.1.3`. Modes: 'ind-cpa' (chaotic nonce) and 'accurate' (zero nonce).

5 Benchmarks

AMD Ryzen 5 2600, 16GB DDR4, Ubuntu 22.04, GCC 11.4.0.

Metric	Value
Throughput (Real encrypt-add-decrypt)	14.8M–15M ops/sec
Ciphertext Size	40 bytes
Noise Floor	40.00 bits
Security Parameter (κ)	256 bits
Effective Key Space (7D CML)	$\sim 2^{420}$
Server Key Knowledge	None
Compiler Warnings	Zero
Dependencies	OpenSSL 3.0+ only

Table 1: Performance and Security Benchmarks

Scheme	TPS	CT	Boot	Blind	Stab	KeyMgmt
FEmmg-FHE v16	15M	40B	None	Yes	Phi-Zeta	Client-only
TFHE	~100	~1KB	Req	No	None	Server
CKKS	~1K	~100KB	Req	No	None	Server
BFV	~100	~100KB	Req	No	None	Server

Table 2: Comprehensive Comparison

5.1 Dark Abyss Gauntlet (30/30)

Basic FHE (5/5), Attack Resistance (5/5), Precision (5/5), Extreme Stress (5/5), Dark Abyss (5/5), Lyapunov+Phi-Zeta (5/5). Total: **30/30**.

5.2 Conjectured φ^n Spacing Verification

Gap #	Actual	$\varphi^n \times 0.134$	Error
0	6.8873	0.1340	6.75
1	3.9888	0.2168	3.77
2	5.4140	0.3508	5.06
3	2.5102	0.5676	1.94
4	4.6511	0.9185	3.73
5	3.3325	1.4861	1.85
6	2.4084	2.4045	0.0038
7	4.6781	3.8906	0.79
8	1.7687	6.2952	4.53

Table 3: Preliminary φ^n Spacing — Gap #6: 99.84%. Conjecture only.

6 Conclusion

FEmmg-FHE v16.0 achieves production-ready Fully Homomorphic Encryption through three-layer stabilization (Banach + Lyapunov + Phi-Zeta). The conjectured φ^n Riemann zero spacing provides a novel attractor for noise management. The two-phase architecture ensures both IND-CPA security and perfect computational accuracy. At 14.8–15M TPS with 40-byte ciphertexts and zero bootstrapping, FEmmg-FHE v16.0 represents a significant advance in practical homomorphic encryption.

Acknowledgments: The φ^n spacing conjecture emerged during FHE noise optimization. The Banach Fixed Point Theorem (1922) and Lyapunov stability (1892) provide the foundation. Bernhard Riemann’s 1859 paper [9] inspired the zeta function application.

Limitations Statement: The φ^n spacing is based on 10 zeros only and is labeled as a conjecture. The Chaotic Trajectory Unpredictability Assumption is novel and unverified by third-party cryptanalysis. The Riemann Hypothesis remains unproven.

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